

# Standardising Meaning, Not Custody

*A research proposition for a music industry ontology*

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## Abstract

The music industry has attempted, on at least four occasions since 1998, to build shared infrastructure for the identification of works and rights holders. Each attempt failed, and each failed for reasons that were organisational rather than technical. This proposal argues that these failures share two causes worth separating. The first is a coordination problem: every design required competing commercial parties to contribute sensitive data to a common pool, and refusal was a rational response rather than an obstruction. The second is a category error: every design addressed identification, which establishes that two records refer to the same entity, and none addressed semantics, which establishes what kind of entity it is, what relationships it participates in, and what operations may validly be performed on it. The industry has therefore standardised how to name things without ever standardising what things are.

This proposal sets out a research programme to define, formally specify, implement and evaluate an ontology for the music industry. Its distinguishing constraint is that the ontology is intended to be common while the data remains separate: the unit of standardisation is the model, and the unit of deployment is an isolated per organisation instance. Because a single deployment is useful to its owner with no other participant involved, the programme does not depend on the industry wide coordination that defeated its predecessors. The proposal also identifies an area that no prior effort has addressed at all, namely the discontinuity between the creation of a recording and its commercial exploitation, and argues that this discontinuity is becoming commercially consequential as provenance and attribution acquire legal and market weight.

**Keywords:** music rights, data infrastructure, ontology, metadata, entity resolution, royalty administration, provenance, attribution

## 1. Introduction

That the music industry has a data problem is not a contested claim. It is one of the few propositions on which record companies, publishers, collecting societies, digital service providers, artist advocates and regulators agree. Royalties go undistributed because works cannot be matched to registrations. Credits are incomplete because no system holds them. Reconciliation consumes permanent headcount at every organisation in the chain, performing work that is structurally identical

from one company to the next. Questions that require joining information from two domains are routinely answered by hand, or abandoned.

Given this consensus, the more interesting question is not whether the problem exists but why it has survived four coordinated attempts to solve it, three of which involved the largest companies in the industry and one of which was convened by this institution. A fifth attempt built on the same assumptions would be difficult to justify. This proposal therefore begins with the failures rather than with the problem, on the view that a diagnosis of why the previous designs collapsed is a precondition for proposing anything new.

The argument advanced here has two parts. The first is that prior initiatives failed on a structural requirement rather than on execution: each asked commercially competing parties to place rights data into a shared pool, which made participation a strategic concession rather than an operational benefit, and which made funding a public goods problem that predictably went unfunded. The second is that all prior initiatives addressed the same layer, identification, and that identification is not the layer at which the industry's difficulty actually resides.

From these two observations the proposal derives a different design: standardise the semantic model and leave custody untouched. What follows sets out the prior work, the diagnosis, the proposition, a reference architecture, the research questions, a method, and the criteria by which the programme should be judged to have succeeded or failed.

## **2. Prior work**

### ***2.1 Four attempts at a shared database***

The International Music Joint Venture, established in 1998 by societies in the United States, the United Kingdom and the Netherlands, was the first serious attempt at a global rights database. It dissolved in 2001 having covered less than a fifth of global repertoire, after labels and publishers declined to participate on the grounds that exposing their repertoire threatened their commercial position (Trabucchi and others, 2023).

The International Music Registry, launched in 2011 under United Nations auspices, collapsed amid disputes over control between labels and publishers (Trabucchi and others, 2023). Its failure is generally attributed to governance rather than to technology.

The Global Repertoire Database is the most instructive case because it came closest and because it is the best documented. Convened through a European Commission working group, it aimed to provide a single authoritative view of works and ownership shares, with International Copyright Enterprise appointed as technology provider and Deloitte as project manager (Cellini, 2026). It was abandoned in July 2014 after roughly eight million pounds had been spent, when participating collecting societies withdrew funding. Public statements at the time reaffirmed commitment to the underlying principles of single point registration and reconciliation of ownership shares while conceding the project could not proceed (Music Ally, 2014; Complete Music Update, 2014). The most careful

account of the failure was published in this institution's own Music Business Journal, which attributed it primarily to the asymmetry between who paid and who benefited (Milosic, 2015).

The Open Music Initiative, founded by the Berklee Institute for Creative Entrepreneurship with the MIT Media Lab in June 2016, differed in convening more than two hundred members, including all three major recorded music companies and the principal digital services, and in pursuing an open source protocol for the uniform identification of rights holders and creators rather than a centrally operated database (Berklee College of Music, 2016). It represents the most broadly supported effort to date and it is the direct institutional predecessor of anything proposed here. Any new work at Berklee on this subject should be understood as continuous with it rather than as a departure from it.

## ***2.2 What has succeeded, and why***

Two categories of effort have succeeded, and the reasons are worth stating precisely because they constrain what a new proposal may reasonably assume.

Interchange standards have succeeded. ISRC, ISWC, IPI and ISNI provide identification. DDEX standardises the messages that pass between trading partners, and CWR moves works registrations between publishers and societies. These are substantial achievements and this proposal depends on them. They succeeded because they require agreement only on the envelope and not on the contents: a company may adopt DDEX without changing anything about how it internally understands a recording, a contract or a party. The cost of adoption is low precisely because the commitment is shallow.

One shared database has succeeded, and it was compelled. The Mechanical Licensing Collective exists because the Music Modernization Act of 2018 created it by statute, funded it by statute, and obliged digital services to use it. It is the exception that clarifies the rule. Where the industry has built shared infrastructure voluntarily, it has failed; where the shared infrastructure was legislated, it exists. A proposal that depends on voluntary industry wide coordination is therefore proposing something for which there is no precedent of success.

## **3. Diagnosis**

### ***3.1 The custody requirement***

Every failed initiative shared a design feature: it required participants to contribute data into a common repository. This is usually described as a coordination problem, which understates it. For a publisher or a label, ownership and share data is not merely sensitive, it is the asset. Contributing it to a pool that competitors and licensees can query is a transfer of commercial position, and the return on that transfer is diffuse and shared with everyone who did not contribute. Declining is not obstruction. It is the correct decision for the party making it.

The funding pattern follows directly. Costs concentrate on whoever volunteers to build, while benefits distribute across the industry including non contributors. In the case of the Global Repertoire

Database, development costs fell on a small number of collecting societies while the beneficiaries would have included every licensee and every rights holder in the market. Withdrawal of funding was the predictable end state of that structure rather than an accident of it.

Any design that begins by asking for data will encounter the same dynamics. This is the first constraint the present proposal accepts.

### ***3.2 The category error***

The second observation is less frequently made. Every initiative described above addressed identification. Identification answers a single question: do these two records refer to the same entity. It is a necessary capability and the industry is right to have invested in it.

But identification presupposes a model and does not supply one. An identifier tells you that two rows concern the same recording. It does not tell you what a recording is, what it may be related to, how it differs from a work or a release or a master, which operations may validly be performed on it, or what must be true for those operations to be legitimate. Two organisations can use ISRC entirely correctly and remain unable to answer a shared question, because one models a recording as a line on a royalty statement, the other as an asset in a distribution system, and neither model contains the concept the question requires.

The industry has a mature standard for how to name things and no standard at all for what things are. That absence is the reason the same reconciliation work is rebuilt at every company, the reason cross domain questions are unanswerable, and the reason institutional knowledge fails to accumulate. It is a semantic problem being treated as an identification problem, and it is the substantive research territory this proposal claims.

## **4. The proposition**

The proposition is that the music industry can be described by a single coherent ontology, that this ontology should function as substrate rather than as output, and that a considerable portion of what the industry currently treats as separate software problems are consequences of its absence.

Four principles follow.

**Objects before applications.** Object types, their relationships and the actions that may be performed on them are defined first, and applications compute against those objects. The prevailing order is the reverse, which is why every application acquires a private model and why no two are compatible.

**Meaning separated from source.** Statement layouts, file structures and partner formats terminate at the ingest boundary and do not propagate into the semantic model. Where they do propagate, every new trading partner becomes a schema migration, which is the ordinary condition of royalty systems today.

**Unresolved states are first class.** In music data, matching is the difficulty rather than an incidental step. A model that represents only successfully resolved entities has modelled the easy half of the

domain. Unmatched identifiers, incomplete share sets and low confidence matches must exist as objects with their own lifecycle and their own queues, not as exceptions absorbed silently by a pipeline. A silent matching failure is a silent financial loss, and the design should make it impossible for one to remain invisible.

**Decisions are data.** When an experienced administrator resolves an ambiguity, that resolution encodes expertise that exists in no other retrievable form. Captured as a typed and immutable event and written back into the model, it allows the system to improve; left uncaptured, it leaves with the individual.

#### ***4.1 The distinguishing constraint***

The design departs from its predecessors in one respect that determines everything else. Each deployment is an isolated instance holding its own ontology, configured for that organisation at deployment. A platform core may be versioned and shared across deployments; the ontology instance, its extensions and the data within it are not. No organisation contributes data to a pool, because no pool exists.

The unit of standardisation is the model. The unit of deployment is the instance.

The consequence is that adoption is not a coordination game. An organisation deploying alone obtains the full operational benefit of a coherent object model over its own data, independent of whether any other organisation ever does the same. Interoperability, if it emerges, emerges because two organisations independently found the same model useful and can therefore exchange meaning rather than merely syntax. Convergence becomes an outcome rather than a precondition, which is the specific respect in which this design is not a fifth attempt at the same thing.

The cost of this constraint should be stated plainly rather than minimised. A design that refuses a shared pool also refuses the network effects that a shared pool would produce, and it cannot deliver a global view of repertoire. What it can deliver is a common language, which is the layer at which the industry has never had a standard, and which does not require anyone to surrender anything to obtain.

### **5. Reference architecture**

The reference implementation is organised as five horizontal layers with three concerns running vertically through all of them.

The first layer performs physical integration: ingest of statements, registrations, catalogues, contracts, product records and telemetry, with source formats terminating at this boundary. The second performs entity resolution, matching on identifiers with explicit confidence scoring and an explicit unresolved surface. The third is the ontology itself, holding object types, relationships, actions and time semantics, and functioning as the substrate against which everything above it computes. The fourth holds applications, which are built against ontology objects rather than against raw data, so that royalty operations, catalogue analysis, credits administration and artist relations are

surfaces on one model rather than separate systems. The fifth performs decision orchestration, where actions are proposed, approved, executed and then written back both into the ontology and into the source systems from which the data came. This write back path is what distinguishes an operational substrate from a reporting tool, and it carries the legal and financial risk of the system, which is why it must be typed, approved, idempotent and fully audited rather than treated as an ordinary write.

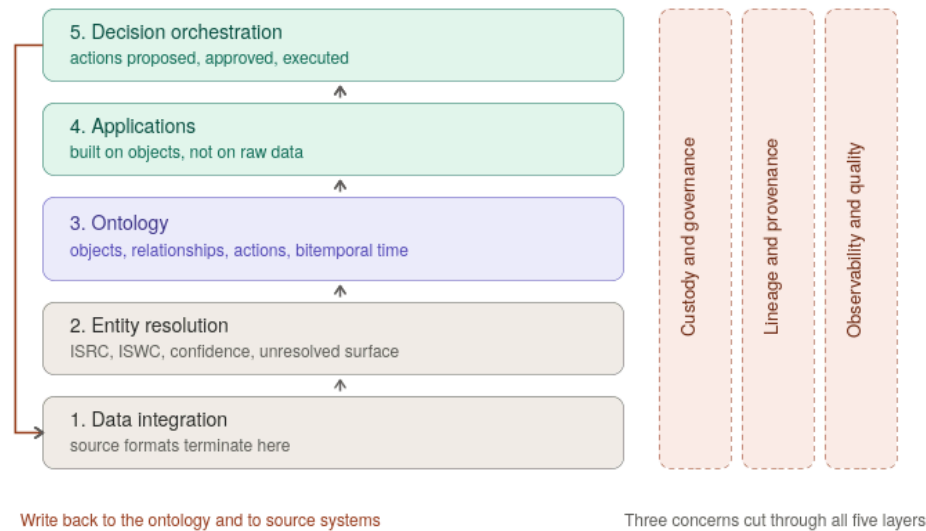


Figure 1. The five layers, with three concerns running vertically through all of them.

Three concerns cannot be located in any single layer and must run through all five. Custody and governance, because the question of who may see what is a precondition of participation rather than a feature. Lineage and provenance, because royalty and credit disputes are adversarial and every figure must be traceable to the document that produced it. Observability and data quality, for the reason given above concerning silent failure.

### 5.1 Modelling commitments requiring research

Three commitments are asserted here on the basis of preliminary work and require empirical validation as part of the programme.

Royalty computation is not a single operation. Preliminary work distinguishes at least three mechanisms with materially different epistemic status: income arising under statutory rates, which can be recomputed from first principles and audited; income arriving at privately negotiated rates, which can only be captured as reported; and income arriving from pooled distributions, which can be reconciled at aggregate level but not recomputed at the level of the individual work. A model that represents these identically produces figures that appear precise and are not. Encoding the distinction explicitly is both a modelling contribution and a safeguard against false confidence in audit contexts.

Contracts are the least modelled and most consequential objects in the domain. Nearly every financial fact in music is downstream of a contractual term, and few systems represent clauses as objects at all. This is, on present evidence, the single largest gap in the domain model.

Time must be bitemporal. Shares are amended, contracts are revised, rates are set retroactively and statements restate. A model that cannot distinguish when a fact was true from when it became known will produce historical analysis that is quietly and irrecoverably wrong.

## 5.2 Relationship to existing systems

The architecture consumes existing industry systems rather than replacing them, and the distinctions between those relationships are material enough to state explicitly.

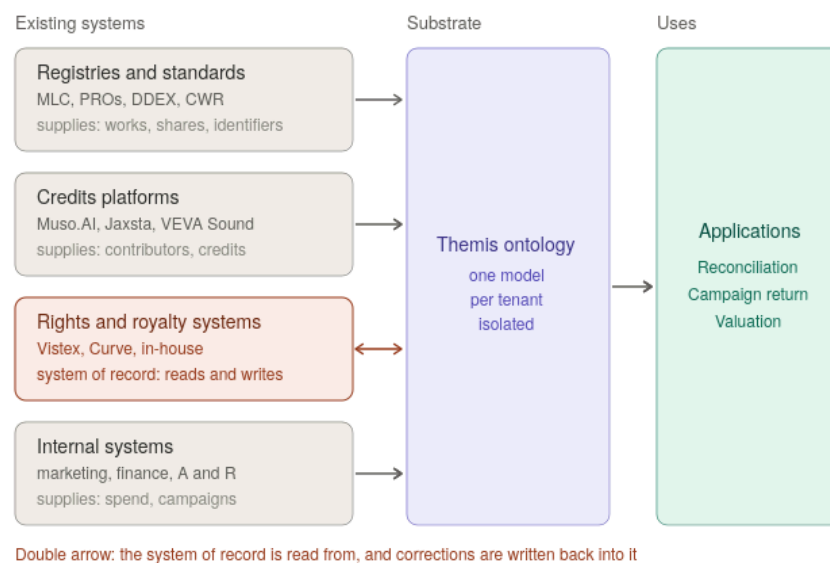


Figure 2. What each category of existing system supplies, and the direction of each relationship.

Credits platforms including Muso.AI, Jaxsta and VEVA Sound, registries and messaging standards including the Mechanical Licensing Collective, the performing rights organisations and DDEX, and internal marketing and finance systems are all sources. Each holds a correct partial model of one part of the domain, and none can see the others. The ontology consumes them, resolves them against one another, and represents what they collectively describe.

Rights and royalty management systems occupy a different position. Established enterprise platforms in this category already model contracts, rates and royalty calculation for a large proportion of the industry, and where such a system is in place it is the organisation's system of record. The relationship is therefore bidirectional: the ontology reads from it, and corrections and decisions must be written back into it rather than held separately. Determining that interface, and deciding what the ontology models independently versus what it defers to the system of record, is among the more consequential design questions the programme would address.

The asymmetry between reading and writing deserves emphasis, because it determines the real cost of deployment. Reading from a system of record is an integration problem. Writing into one is a permissions, approval, audit and liability problem, since an incorrect write does not produce a misleading report but corrupts the accounting record from which payments and statements are generated. The write path must therefore be typed, explicitly approved, idempotent and fully audited, and treated as a distinct engineering and governance concern rather than as an ordinary database operation. This is a principal reason deployments of this kind are slow, and it is also why such a write path is difficult to displace once established, since removing it requires unwinding every decision that has passed through it.

The horizontal claim is visible in the same figure. If the deploying organisation is a hardware manufacturer or a talent agency rather than a label, the left column changes entirely while the ontology and the applications beside it retain their shape. The substrate is common; the sources and the first application are not.

## **6. Cross domain measurement: the case of promotional spend**

The preceding sections argue the general case. This section takes a single instance and works it through, both because it is the clearest illustration of a question that cannot be answered without a common model, and because it concerns the largest discretionary expenditure a recorded music company makes. The question is simple to state and, at present, cannot be answered: what did this campaign earn?

The obstacle is that marketing operates in consumption measures and the company earns in money, and the two are not convertible without information marketing does not hold. A campaign is evaluated on streams, saves, followers, playlist placement and chart position. None of these is revenue. What a stream is worth depends on the platform it occurred on, the territory, whether the listener was a subscriber or on an advertising supported tier, and the terms of the specific agreement governing that recording. The same stream count can differ in value by a large multiple depending on those variables, all of which are held on the royalty side of the company. Marketing therefore optimises a proxy, and the relationship between that proxy and income is assumed rather than measured.

Three consequences follow, in increasing order of seriousness.

Campaign return cannot be computed. Expenditure is recorded in one system and income in another, with no shared notion of the recording, the territory or the period that would allow them to be joined. A label can state what a campaign cost and approximately what it moved in consumption terms. It cannot state what it earned, which means the largest discretionary budget in the business is allocated substantially on judgement and precedent.

Recoupment is invisible to the people affecting it. Promotional expenditure is commonly recoupable in part against artist royalties, the proportion varying by agreement. Marketing decisions therefore alter directly what an artist is paid and when they begin to be paid, while the people making those

decisions generally have no view of the recoupment balance they are moving. This is not only an efficiency problem. It is a fairness problem, and it is one that artists increasingly raise.

Revenue quality is indistinguishable from revenue volume. Promotion that drives consumption in high rate territories, on subscription tiers, or on recordings the company owns outright is worth considerably more than promotion producing identical consumption on low rate advertising supported tiers or on recordings held under short licence. Nothing in the current arrangement communicates that difference back to the people allocating spend, so the feedback loop that would teach it never closes.

It is worth noting that the public accounts do not pair the two either. Warner Music Group defines its artist and repertoire cost line as covering royalties paid to recording artists, producers, songwriters and other rights holders, the signing and development of artists, and the creation of master recordings (Rolling Stone, 2020). Promotional expenditure is not presented alongside it as a comparable line. The categories that external reporting uses do not correspond to the decisions the business actually makes, which is a smaller version of the same problem.

What answering the question requires is not new data. Campaign records, expenditure, consumption reporting and royalty income all already exist inside the company. What is missing is a model in which a campaign, a recording, a territory, a period and an agreement mean the same thing across all of them, so the join is possible at all. That is precisely the semantic layer described in section four, applied to one question.

The example also demonstrates the point made in section 4.1 concerning deployment. Nothing here requires the cooperation of another company. The data is entirely internal, the beneficiary and the payer are the same organisation, and the funding asymmetry that defeated earlier industry initiatives does not arise. A single deployment answers the question for the company that deploys it.

The evidential status of this section should be stated honestly. A preliminary conversation with a contact at one major recorded music company indicated that the company operates numerous unconnected in house systems, that departments are weakly connected to one another, and that marketing shares this condition. That is a single account from a single organisation and it is treated here as a lead rather than a finding. Establishing whether the pattern generalises, and quantifying what it costs, is the substance of RQ2 and among the first tasks the programme would undertake.

One framing caution follows from the same evidence. The requirement is not that a royalty department be connected to a marketing department, which describes plumbing and interests nobody. The requirement is an answer to a question that a finance director already wants answered and which happens to need both.

## **7. The creation and exploitation discontinuity**

The preceding sections concern territory that prior initiatives approached and failed to complete. This section concerns territory that no prior initiative has approached at all, and it is where the proposal makes its strongest claim to novelty.

A recording exists twice in the world and the two versions are never connected. On one side it is a rights object: an identifier, a set of shares, a chain of agreements, a history of income. On the other it is a production artifact: sessions, performers, instruments, signal chains, rooms, engineers, software, and a sequence of decisions that produced the final master. No system in the industry can traverse from one representation to the other. A recording cannot be traced backwards to how it was made, and a piece of equipment or a session cannot be traced forwards to what was made with it.

The two halves are held by different constituencies. Record companies and publishers hold the exploitation side and retain almost no structured record of production. Instrument and audio hardware manufacturers, together with software instrument and recording tool vendors, sit closest to the creation side and have no access whatever to the exploitation side. Both are impoverished by the discontinuity, and neither can repair it alone.

Four consequences are worth setting out, because they establish that this is a present commercial problem rather than an archival curiosity.

Artist relations cannot be measured. Endorsement and artist relations programmes are among the largest marketing commitments made by instrument and audio companies, and they are evaluated almost entirely on judgement, because no mechanism exists to connect a relationship with an artist to the works that artist produced or to the commercial performance of those works. This is a measurement problem produced by a missing object model rather than by missing data.

Product identity and creator identity are disconnected. Manufacturers increasingly hold registration records, account data, software licences and in some cases usage telemetry. None of it connects to any external notion of a creator or a work, with the result that a company may know a device is in active use and know nothing about what is being made with it.

Credits and attribution remain unsolved. Performers, engineers and producers are systematically under credited, and credits are not merely a matter of recognition: they drive registrations, royalty entitlements and professional reputation. The production lineage that manufacturers sit closest to is precisely the information that would address this, and there is at present no model into which it could be recorded.

Provenance is acquiring commercial and legal weight. As synthetic audio becomes ordinary, the ability to demonstrate how a recording was actually made, on what equipment, by whom and in what sequence, moves from being a matter of documentation to being an assertion with market and evidentiary consequences. Establishing provenance requires an object model of production. No standardised model exists, and the parties best positioned to assert one have no framework in which to do so.

The research contribution proposed here is to extend the ontology across this discontinuity, so that production objects and rights objects are expressible in a single model even while remaining in entirely separate custody. The per instance deployment constraint is what makes this tractable: a manufacturer and a label may share a model without either disclosing anything to the other.

## **8. Research questions**

**RQ1.** What are the irreducible object types, relationships and actions of the music industry, and can a single ontology span creation, rights and consumption without collapsing either into an abstraction too general to be useful or into a set of disconnected vertical models?

**RQ2.** Where does the object graph break in practice, and what is the measurable cost of each break? This converts an architectural claim into an empirical one and is the question most likely to produce findings of independent value.

**RQ3.** Given that participants will not contribute to a shared pool, what deployment model permits semantics to be common while custody remains separate, and what is forgone by accepting that constraint?

**RQ4.** What becomes computable once the ontology exists that is not computable today, and by what margin does it improve on current practice? This is the evaluation question and the one that determines whether the thesis is correct.

## **9. Method**

The programme is proposed in five phases.

The first phase is elicitation. The object model is reconstructed domain by domain from primary material, meaning actual contracts, actual royalty statements, actual registration files, actual session and product records, supplemented by structured interviews with practitioners. The ontology is derived from documents rather than from opinion, since a model built from what practitioners say they do reliably diverges from what their paperwork shows they do. Preliminary elicitation has already produced a working model of the operational domains of a major recorded music company, comprising several hundred object types and action types across approximately two dozen functional departments. This is a starting point for validation, not a result.

The second phase is formal specification: object types, relationships, action types, confidence and resolution states, and bitemporal semantics, specified in sufficient detail that a third party could implement it independently. This specification is the principal publishable artifact of the programme.

The third phase is reference implementation: a working substrate implementing the specification, deployed for each partner on that partner's own data and within that partner's custody. Substantial engineering already exists, including ingest with deterministic idempotency, royalty line persistence, identifier resolution with an explicit unresolved surface, catalogue ingest with audited re resolution,

and contract, clause and share ingest with incomplete share sets flagged rather than silently completed.

The fourth phase is evaluation against the criteria set out below.

The fifth phase is publication. The specification is proposed for open release. Implementations, including the author's, remain the property of those who build them. The ontology is proposed as a public good; the substrate is not.

## **10. Evaluation**

The programme should be assessed against measurable outcomes rather than against the plausibility of its architecture. Five criteria are proposed.

Coverage. What proportion of real documents drawn from a partner's actual operations map into the ontology without requiring extension. A model that needs a new object type for each new contract is not a model.

Reconciliation accuracy. Whether recomputed royalty figures tie exactly to actual statements across each of the three income mechanisms, with every discrepancy explained rather than absorbed. This is a pass or fail test requiring no interpretation, it can be verified by a partner against their own records, and it should be run first.

Extension cost. How much work is required to onboard an additional partner or source format once the core exists. If this figure does not fall sharply after the first several deployments, the separation of source from semantics has failed and the thesis is weakened.

Cross domain capability. A fixed set of questions that cannot be answered today, answered from the substrate, with results verified against manual reconstruction.

Decision capture. Whether human resolutions are recorded, retained and demonstrably improve subsequent resolution rates, which tests whether the write back path compounds or merely logs.

## **11. Scope**

Four exclusions are stated to prevent predictable misreadings.

This is not a catalogue valuation or market intelligence product. Valuation is one application among many and does not define the substrate.

This is not an analytics or dashboard tool. Reading data is a commodity capability. The proposition concerns the model beneath it and the ability to act upon it.

This is not a shared industry database, a rights registry, or a distributed ledger. The last of these deserves explicit exclusion given the period in which the preceding initiative was conceived.

Distributed ledger designs address trust between parties who share a record. The present design has no shared record, so the problem those systems solve does not arise.

This is not a replacement for existing identifier or messaging standards. It consumes them and depends upon them, and addresses the layer above them that they were never intended to occupy.

## **12. Limitations and risks**

Access to data is the binding constraint. The research depends on real operational data from partners in both constituencies, and without it the ontology becomes speculative, which is the failure mode of committee designed standards. The mitigation is that deployment occurs within the partner's own custody and that access begins with the least sensitive domains, but the risk cannot be eliminated by design alone and will require institutional convening to manage.

The precedent is discouraging. Four attempts have failed and a fifth proposal invites reasonable scepticism. The response offered here is not confidence but structure: because a single deployment is useful in isolation, the programme does not require the industry wide agreement that defeated its predecessors, and it can therefore fail partially rather than totally.

Scope is unbounded in principle. An ontology of an entire industry can expand without limit. The mitigation is that scope is bounded empirically by coverage testing against real documents rather than by ambition, and that the model is versioned and extended per deployment rather than being completed in advance.

Semantic drift across deployments is the cost of the central design decision. If each instance extends its ontology freely, the common model erodes and the standardisation claim weakens. A versioned core with a formal extension mechanism is the intended mitigation, but the tension between local fit and shared meaning is genuine and is itself among the more interesting questions the programme would investigate.

Finally, the creation side argument set out in section seven is the least evidenced part of this proposal. It rests on an assessment of where commercial pressure is moving rather than on demonstrated demand, and it should be tested early with manufacturers directly rather than assumed.

## **13. Institutional requirements**

The programme requires three things that an institution can provide and a commercial entity generally cannot.

The first is convening authority. Access to record companies, publishers, manufacturers and standards bodies in a research capacity is a different conversation from a commercial one, and it opens doors that a vendor relationship does not. Berklee has demonstrated this capacity before, having assembled more than two hundred organisations including all three major recorded music companies around a related question in 2016.

The second is data partnership. Formal agreements permitting work on real operational data under appropriate confidentiality, without which the elicitation phase cannot proceed honestly.

The third is academic supervision and research capacity. Faculty oversight, and graduate researchers for elicitation, document analysis and evaluation, which is work well suited to research training and which produces publishable output.

In exchange the programme would produce an openly published specification, a body of empirical findings concerning where the industry's data infrastructure fails and at what cost, a working reference implementation, and an institutional position in a standards conversation that will proceed with or without academic participation.

## **14. Conclusion**

The music industry has spent close to three decades solving the problem of naming things and has not begun the problem of describing them. The difficulties the industry treats as permanent operating costs, meaning reconciliation, unmatched income, absent credits, unanswerable questions and unverifiable provenance, are downstream of that omission rather than independent of it.

The proposition is not that the industry needs a better application. It is that applications have been built on an absent foundation, that four attempts to supply one failed because they asked for data instead of asking for agreement about meaning, and that the absent foundation is the research problem worth taking up.

## **Disclosure**

The author is developing a reference implementation of the architecture described in section five through Aegis Technologies. The specification described in section nine is proposed for open publication; implementations, including the author's, would remain the property of those who build them. This disclosure is made so that any institutional engagement proceeds with full knowledge of the author's commercial interest, and the governance of any resulting arrangement is properly a matter for the institution to determine.

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